Fiscal Year:	FY 2011	**Task Last Updated:**	FY 04/12/2011
PI Name:	Chang, Polly Y. Ph.D.		
Project Title:	Tissue-specific acute and late molecular surveillance of particle radiation effects		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation Biology		
Joint Agency Name:	**TechPort:**		No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	polly.chang@sri.com	**Fax:**	FY 650-859-2889
PI Organization Type:	NON-PROFIT	**Phone:**	650-859-2549
Organization Name:	SRI International		
PI Address 1:	Biosciences, PN 187		
PI Address 2:	333 Ravenswood Avenue		
PI Web Page:			
City:	Menlo Park	**State:**	CA
Zip Code:	94025-3453	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2007 Space Radiation NNJ07ZSA001N
Start Date:	11/01/2007	**End Date:**	10/31/2010
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
Contact Email:	noaccess@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX07AV20G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

The space atmosphere is comprised of wide spectrum of highly energetic and charged particle radiation. Exposure to this type of radiation in the galactocosmic environment or unplanned solar particle events (SPE) may pose a potential health hazard to humans in long-term manned space explorations. A large database of information is available on the mutagenic potential of particle radiation on cultured cells, however, limited information is available for multicellular tissue responses. This is primarily due to constraints in the selection of adequate markers in the appropriate tissue systems. Transgenic rodent mutation model systems are widely accepted by regulatory agencies as an appropriate in vivo systems to access the potential carcinogenicity of a variety of test articles. In these systems, multiple copies of the inert reporter gene are integrated into every cell of the genome, making it possible to use it as a tool to measure, both quantitatively and qualitatively, mutations in every tissue of the animal. In our studies, we are using one of the transgenic mouse mutation model systems to evaluate the immediate and long-term mutagenic potential of simulated space radiation beams in different tissue. Knowledge gained in this ground-based work is directed at understanding how tissue and organ respond to exposures both on the lunar surface as well as in deep space. Information gathered from this research will aid in reducing the uncertainties associated with predicting risks both in the estimation of risk to astronaut health and the design of appropriate countermeasures to ameliorate any potential negative effects in space flights.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

The space atmosphere is comprised of particle radiation in a wide spectrum of highly energetic and charged (HZE). Exposures to these HZE radiations poses a serious health hazard to humans in long-term manned space explorations. We are using the plasmid-based lacZ transgenic mouse mutation model system to examine the acute and long-term tissue specific mutagenic potential after exposure to radiation with different qualities. Knowledge gained in this ground-based work is directed at understanding tissue and organ response to space radiation and will aid both in the estimation of risk to astronaut health and the design of appropriate countermeasures to ameliorate the negative effects in space flights.

Task Progress:

[Editor's note 4/12/2011: No Task Book report received. Progress section and Bibliography compiled, after unsuccessful attempts to receive direct reporting, from PI's Final Technical Report.]
We have demonstrated in our current project that the lacZ model system is indeed an efficient and effective model system that can be used to evaluate tissue-specific genomic damage as a function of particle radiation quality. In addition to measuring reporter transgene modifications, we have taken advantage of the in vivo aspect of the system by using a multi-pronged approach to evaluate particle radiation induced perturbations in different organ systems, including cytogenetic damage and recovery in reticulocytes (RETs) populations and lymphocytes and alterations in gene expression profiles in the CNS and the liver tissue. Through this effort, we have identified temporally-regulated dose, dose-rate and radiation quality- and dosing regimen-dependent of changes in (i) mutation in the reporter transgene, (ii) gene expression profiles in micro-dissected hippocampus sections from the brain and liver tissues, (iii) persistence of total chromosome/chromatid aberrations as well as chromosome-specific lesions, and (iv) dynamic early changes in levels of RETs and MN-RETs in peripheral blood. Mutation analysis of two different tissues in the same animals highlights the importance of tissue specificity considerations in risk analysis. Sequential exposure to particle radiations with different LETs resulted in alterations in gene expressions profiles that are distinctly different from single ion exposures.

Our results to date suggest that low dose rate sSPE radiation impacts the hematopoietic system. However, such events are transient. On the other hand, gene expression profiling in the hippocampus demonstrated that sSPE induced down regulation of neurotrophic genes important to functional integrity of CNS. We believe that the concept of dose response must be expanded to not only include quantitative changes in the level of response, but also qualitative changes in the profile of genetic expression. Obviously, higher doses of sSPE displays a gene expression spectrum that is qualitatively different from those obtained from animals that were exposed to lower doses.

Bibliography Type: Description: (Last Updated: 04/26/2023)

Abstracts for Journals and Proceedings	Doppalapudi R, Davis Z, Menda S, Bakke J, Chang P. "Comprehensive Evaluation of LET-dependence of Chromosome Aberrations after Particle Radiation Exposure in vivo." 41st Environmental Mutagen Society Annual Meeting, Fort Worth, Texas, Oct 23 – 27, 2010. Environmental and Molecular Mutagenesis 2010 Aug;51(7):669-742. http://dx.doi.org/10.1002/em.20624 , Aug-2010
Abstracts for Journals and Proceedings	Chang PY, Bakke J, Wang A. "Impact of sequential exposure to protons and iron ions in vivo." 21st Annual NASA Space Radiation Investigators' Workshop, Port Jefferson, NY, May 16-19, 2010. Program and abstracts. 21st Annual NASA Space Radiation Investigators' Workshop, Port Jefferson, NY, May 16-19, 2010. p. 28. , May-2010
Abstracts for Journals and Proceedings	Chang PY, Bakke J, Wang A. "Impact of sequential exposure to protons and iron ions in vivo." Presented at the 56th Annual Meeting of Radiation Research Society, Maui, HI, September 25-29, 2010. 56th Annual Meeting, Radiation Research Society, September 25-29, 2010. http://www.abstractsonline.com/Plan/ViewAbstract.aspx?mID=2569&sKey=359561f3-493c-4d5c-b014-47e677719c70&cKey=459eb20a-3ca2-4a69-84b6-a5980e5be9bb&mKey=5fb93787-2553-4b3b-a915-8c1ac71f127a , Sep-2010
Articles in Peer-reviewed Journals	Chang PY, Doppalapudi R, Bakke J, Wang A, Menda S, Davis Z. "Biological impact of low dose-rate simulated solar particle event radiation in vivo." Radiat Environ Biophys. 2010 Aug;49(3):379-88. Epub 2010 May 16. PMID: 20473680 , Aug-2010